

Tomcat

```
#!/bin/bash
# =====
# ■■■4■Tomcat■■■■■■■■■■
# ■■■■■■Tomcat■■■■■■■■■■■■■■■■■■■■
# ■■■■■■■■
# ■■■■20231035109
# =====
# ■■■■■■
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
BLUE='\033[0;34m'
CYAN='\033[0;36m'
NC='\033[0m'
# Tomcat■■■■■
TOMCAT_VERSION="10.1.18"
TOMCAT_INSTALL_DIR="/opt"
TOMCAT_HOME="${TOMCAT_INSTALL_DIR}/tomcat-${TOMCAT_VERSION}"
TOMCAT_DOWNLOAD_URL="https://dlcdn.apache.org/tomcat/tomcat-10/v${TOMCAT_VERSION}/bin/apache-tomcat-${TOMCAT_VERSION}.tar.gz"
TOMCAT_ARCHIVE="apache-tomcat-${TOMCAT_VERSION}.tar.gz"
TOMCAT_LOG="/workspace/linux_course_design/tomcat_install.log"
# JDK■■■■■Tomcat■■■■Java■■■■
JAVA_HOME=""
JDK_VERSION="11"
# ■■■■■■
init_tomcat_log() {
echo "===== " > "$TOMCAT_LOG"
echo "Tomcat■■■■■" >> "$TOMCAT_LOG"
echo "■■■■■: $(date '+%Y-%m-%d %H:%M:%S') " >> "$TOMCAT_LOG"
echo "===== " >> "$TOMCAT_LOG"
}
tomcat_log() {
local msg="$1"
local timestamp=$(date '+%Y-%m-%d %H:%M:%S')
echo "[${timestamp}] $msg" | tee -a "$TOMCAT_LOG"
}
# =====
# ■■■1■■■■Java■■■
# ■■■■■■
# 1. Tomcat■■■■■Java■■■■JDK/JRE■
# 2. ■■■■■■■■■■Java
# 3. ■■■■■■■■■■OpenJDK
# 4. ■■■JAVA_HOME■■■■■
# =====
check_java_environment() {
echo -e "${CYAN}= ${NC}"
echo -e "${CYAN} Java■■■■■${NC}"
echo -e "${CYAN}= ${NC}"
tomcat_log "[■■■1] ■■■Java■■■..."
# 1.1 ■■■■■■■■Java
echo -e "${YELLOW}[■■■] Java■■■■■...${NC}"
if command -v java &>/dev/null; then
JAVA_VERSION=$(java -version 2>&1 | head -n 1)
echo -e "${GREEN}■■■■: $JAVA_VERSION${NC}"
# ■■■JAVA_HOME■■■
if [ -n "$JAVA_HOME" ]; then
echo -e "${GREEN} JAVA_HOME: $JAVA_HOMES${NC}"
else
# ■■■■■■■■JAVA_HOME
JAVA_HOME=$(dirname $(dirname $(readlink -f $(which java))))
echo -e "${YELLOW}■■■■■JAVA_HOME: $JAVA_HOMES${NC}"
fi
tomcat_log "Java■■■■: $JAVA_VERSION"
return 0
fi
echo -e "${RED}■■■■Java■■■■${NC}"
tomcat_log "■■■■Java■■■■■■■■■■..."
# 1.2 ■■■Java■■■■■■■■■■■■■■■■■■■■
read -p "■■■■■■■■■■OpenJDK(y/n): " install_jdk
if [ "$install_jdk" = "y" ] || [ "$install_jdk" = "Y" ]; then
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install_java
else
echo -e "${RED}[██] Tomcat██Java██████████${NC}"
return 1
fi
}
# ██Java██
install_java() {
echo -e "${YELLOW}[██] OpenJDK...${NC}"
if [ -f /etc/os-release ]; then
. /etc/os-release
fi
case "$OS_NAME" in
*"Ubuntu"*|"Debian"*)
sudo apt-get update -y
sudo apt-get install -y openjdk-${JDK_VERSION}-jdk
;;
*"CentOS"*|"Red Hat"*|"Rocky"*|"AlmaLinux"*)
sudo yum install -y java-${JDK_VERSION}-openjdk-devel
;;
*)
echo -e "${RED}[██] ██████████${NC}"
return 1
;;
esac
if [ $? -eq 0 ]; then
# ██JAVA_HOME
export JAVA_HOME=$(dirname $(dirname $(readlink -f $(which java))))
echo -e "${GREEN}[██] Java██████: $(java -version 2>&1 | head -n 1)${NC}"
tomcat_log "Java██████, JAVA_HOME=$JAVA_HOME"
return 0
else
echo -e "${RED}[██] Java██████${NC}"
return 1
fi
}
# =====
# ██2██Tomcat
# ████████
# 1. █Apache██████████Tomcat
# 2. ████wget██curl██████
# 3. ████MD5/SHA███
# 4. ████████
# =====
download_tomcat() {
echo -e "${CYAN}=====${NC}"
echo -e "${CYAN}██Tomcat${NC}"
echo -e "${CYAN}=====${NC}"
tomcat_log "[██2] ██Tomcat ${TOMCAT_VERSION}..."
# 2.1 ████████
if [ -d "$TOMCAT_HOME" ]; then
echo -e "${YELLOW}Tomcat██████████: $TOMCAT_HOME${NC}"
read -p "██████████(y/n): " redownload
if [ "$redownload" != "y" ] && [ "$redownload" != "Y" ]; then
echo -e "${YELLOW}██████████${NC}"
return 0
fi
# ████████
sudo rm -rf "$TOMCAT_HOME"
fi
# 2.2 ████████
TMP_DIR="/tmp/tomcat_install"
mkdir -p "$TMP_DIR"
cd "$TMP_DIR"
# ████████
rm -f "$TOMCAT_ARCHIVE"
echo -e "${YELLOW}[██] Apache Tomcat ${TOMCAT_VERSION}...${NC}"
echo -e "███: $TOMCAT_DOWNLOAD_URL"
echo ""
# 2.3 ███wget██curl██
DOWNLOAD_SUCCESS=0
if command -v wget >/dev/null; then
echo -e "${YELLOW}██ wget ██...${NC}"
wget --progress=bar:force "$TOMCAT_DOWNLOAD_URL" -O "$TOMCAT_ARCHIVE" && \

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[illegible]

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fi
echo -e "\n${GREEN}[■■■] Tomcat■■■■■${NC}"
echo -e " ■■■■: $TOMCAT_HOME"
echo -e " ■■■■: /opt/tomcat"
tomcat_log "Tomcat■■■■: $TOMCAT_HOME"
# ■■■■■■
echo -e "\n${YELLOW}--- Tomcat■■■■ ---${NC}"
ls -la "$TOMCAT_HOME/" | head -15
}
# =====
# ■■4■■■Tomcat■■■■
# ■■■■
# 1. ■■CATALINA_HOME■■■■Tomcat■■■■
# 2. ■■PATH■■■Tomcat■bin■■
# 3. ■■■■■■■■■■■■■■
# 4. ■■Tomcat■■■■■■■■
# =====
configure_tomcat_env() {
echo -e "${CYAN}===== ${NC}"
echo -e "${CYAN} ■■Tomcat■■■${NC}"
echo -e "${CYAN}===== ${NC}"
tomcat_log "[■■4] ■■Tomcat■■■■..."
# 4.1 ■■JAVA_HOME■■■■■■■■
if [ -z "$JAVA_HOME" ]; then
if command -v java &>/dev/null; then
export JAVA_HOME=$(dirname $(dirname $(readlink -f $(which java))))
else
echo -e "${RED}[■■] ■■■■Java■■■${NC}"
return 1
fi
fi
# 4.2 ■■Tomcat■■■■■■■■
ENV_FILE="/etc/profile.d/tomcat.sh"
echo -e "${YELLOW}[■■] ■■■■■■...${NC}"
sudo tee "$ENV_FILE" > /dev/null << EOF
#!/bin/bash
# Tomcat■■■■■■
# ■■■■■■: $(date '+%Y-%m-%d %H:%M:%S')
# Tomcat■■■■CATALINA_HOME■Tomcat■■■■■■■■
export CATALINA_HOME=$TOMCAT_HOME
export TOMCAT_HOME=$TOMCAT_HOME
# ■Tomcat bin■■■■■■PATH■■■■■■startup.sh■■■■
export PATH=$PATH:$TOMCAT_HOME/bin
# Java■■■■
export JAVA_HOME=$JAVA_HOME
EOF
sudo chmod +x "$ENV_FILE"
# ■■■■■■
source "$ENV_FILE"
export CATALINA_HOME=$TOMCAT_HOME
export PATH=$PATH:$TOMCAT_HOME/bin
echo -e "${GREEN}[■■] ■■■■■■...${NC}"
echo -e " CATALINA_HOME=$TOMCAT_HOME"
echo -e " ■■■■: $ENV_FILE"
tomcat_log "■■■■■■■■: CATALINA_HOME=$TOMCAT_HOME"
# 4.3 ■■Tomcat■■■■■■
configure_tomcat_settings
}
# ■■Tomcat■■■■■■
configure_tomcat_settings() {
echo -e "\n${YELLOW}[■■] Tomcat■■■■■...${NC}"
SERVER_XML="$TOMCAT_HOME/conf/server.xml"
if [ ! -f "$SERVER_XML" ]; then
echo -e "${RED}[■■] ■■■server.xml■■■■${NC}"
return 1
fi
# ■■■■
cp "$SERVER_XML" "${SERVER_XML}.bak"
echo -e " ■■■■■■: ${SERVER_XML}.bak"
# ■■■■■■■■8080■■■■■■■■80■
read -p "■■■■Tomcat■■■■8080■(y/n): " change_port
if [ "$change_port" = "y" ] || [ "$change_port" = "Y" ]; then
read -p "■■■■■■■ (■■8080/80/8888): " new_port
if [ -n "$new_port" ] && [[ "$new_port" =~ ^[0-9]+$ ]]; then

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# sed -i "s/port=\\"8080\\"/port=\\"$new_port\\"/" "$SERVER_XML"
sed -i "s/port=\\"8080\\"/port=\\"$new_port\\"/" "$SERVER_XML"
echo -e "${GREEN} ████████: $new_port${NC}"
TOMCAT_PORT=$new_port
fi
else
TOMCAT_PORT=8080
fi
# UTF-8
sed -i 's/URIEncoding="[^"]*" /URIEncoding="UTF-8"/g' "$SERVER_XML"
echo -e " URI ████████UTF-8"
echo -e "${GREEN}[██] Tomcat██████${NC}"
tomcat_log "Tomcat██████, ██=${TOMCAT_PORT:-8080}"
}
# =====
# 5███Tomcat██
# 1.███
# 2.███startup.sh███
# 3.███
# 4.███
# =====
start_tomcat() {
echo -e "${CYAN}===== ${NC}"
echo -e "${CYAN}███Tomcat███${NC}"
echo -e "${CYAN}===== ${NC}"
tomcat_log "[██5]███Tomcat..."
# 5.1███Tomcat██████
if [ ! -d "$TOMCAT_HOME" ]; then
echo -e "${RED}[██] Tomcat██████████████████${NC}"
return 1
fi
# 5.2██████
if pgrep -f "catalina" > /dev/null || pgrep -f "tomcat" > /dev/null; then
echo -e "${YELLOW}[██] Tomcat██████████${NC}"
read -p "██████Tomcat(y/n): " restart_choice
if [ "$restart_choice" = "y" ] || [ "$restart_choice" = "Y" ]; then
stop_tomcat_silent
else
echo -e "${YELLOW}██████${NC}"
return 0
fi
fi
# 5.3███Tomcat
echo -e "${YELLOW}[██] Tomcat██████...${NC}"
cd "$TOMCAT_HOME"
# ███JAVA_HOME███
export JAVA_HOME="$JAVA_HOME:-$(dirname $(dirname $(readlink -f $(which java))))"
export CATALINA_HOME="$TOMCAT_HOME"
# ██████████
./bin/startup.sh
if [ $? -eq 0 ]; then
echo ""
echo -e "${GREEN}[██] Tomcat██████████${NC}"
# 5.4██████
echo -e "${YELLOW}[██] ████████...${NC}"
sleep 5
# 5.5██████
check_tomcat_status
else
echo -e "${RED}[██] Tomcat██████${NC}"
echo -e "${YELLOW}██████: $TOMCAT/logs/catalina.out${NC}"
tomcat_log "Tomcat██████"
return 1
fi
}
# ███Tomcat██████████████████
stop_tomcat_silent() {
cd "$TOMCAT_HOME"
export JAVA_HOME="$JAVA_HOME:-$(dirname $(dirname $(readlink -f $(which java))))"
./bin/shutdown.sh 2>/dev/null
sleep 2
# ██████████
pkill -9 -f catalina 2>/dev/null
```

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}
# =====
# ■■■6■■■Tomcat■■■
# ■■■
# 1. ■■■Tomcat■■■
# 2. ■■■
# 3. ■■■HTTP■■■
# 4. ■■■
# =====
check_tomcat_status() {
echo -e "\n${YELLOW}--- Tomcat■■■■■ ---${NC}\n"
PORT=${TOMCAT_PORT:-8080}
STATUS_OK=true
# 6.1 ■■■
echo -e "[■■] ■■■..."
if pgrep -f ".*catalina.*$TOMCAT_HOME" > /dev/null || \
pgrep -f ".*org.apache.catalina.startup.Bootstrap" > /dev/null; then
PID=$(pgrep -f ".*org.apache.catalina.startup.Bootstrap" | head -1)
echo -e " ${GREEN}● ■■■ (PID: $PID)${NC}"
else
echo -e " ${RED}■ ■■■${NC}"
STATUS_OK=false
fi
# 6.2 ■■■
echo -e "\n[■■] ■■■..."
if netstat -tlnp 2>/dev/null | grep -q ":${PORT}" || \
ss -tlnp 2>/dev/null | grep -q ":${PORT}"; then
echo -e " ${GREEN}● ■■ $PORT ■■■${NC}"
else
echo -e " ${RED}■ ■■ $PORT ■■■${NC}"
STATUS_OK=false
fi
# 6.3 HTTP■■■
echo -e "\n[■■] HTTP■■■..."
if command -v curl &>/dev/null; then
HTTP_CODE=$(curl -s -o /dev/null -w "%{http_code}" http://localhost:$PORT/ --connect-timeout 5
2>/dev/null)
case $HTTP_CODE in
200|302)
echo -e " ${GREEN}● HTTP■■■ (■■■: $HTTP_CODE)${NC}"
;;
000)
echo -e " ${RED}■ ■■■${NC}"
STATUS_OK=false
;;
*)
echo -e " ${YELLOW}■ HTTP■■■: $HTTP_CODE${NC}"
;;
esac
fi
# 6.4 ■■■
echo ""
if [ "$STATUS_OK" = true ]; then
echo -e "${GREEN}=====${NC}"
echo -e "${GREEN} Tomcat■■■■■${NC}"
echo -e "${GREEN} ■■■: http://localhost:$PORT${NC}"
echo -e "${GREEN}=====${NC}"
tomcat_log "Tomcat■■■■, ■■=http://localhost:$PORT"
return 0
else
echo -e "${RED}=====${NC}"
echo -e "${RED} Tomcat■■■■■${NC}"
echo -e "${RED} ■■■: $TOMCAT_HOME/logs/catalina.out${NC}"
echo -e "${RED}=====${NC}"
tomcat_log "■■: Tomcat■■■■"
return 1
fi
}
# =====
# ■■■7■■■Tomcat■■■
# =====
stop_tomcat() {
echo -e "${CYAN}=====${NC}"
echo -e "${CYAN} ■■Tomcat■■■${NC}"

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[illegible]

```
configure_tomcat_env && \  
start_tomcat  
echo -e "${GREEN}[██] ██████████${NC}"  
;;  
0)  
echo -e "${YELLOW}████Tomcat██████████${NC}"  
exit 0  
;;  
*)  
echo -e "${RED}[██] ████████████████████${NC}"  
;;  
esac  
echo ""  
read -p "████████..."  
done  
}  
# █████  
main_menu
```